

A₇: new bounds and harmonic Hersch

Answers to Round 5

Continuation at [1ef3f6f8262331d61a84d05e0265658f7ecb7f67](#)

Scope. This note contains the new arguments needed for Round 5. All algebraic curves are smooth, connected and projective over $\mathbb{C}$; actions on them are faithful. A moving bundle is genuinely group-linearized, base-point-free and has at least two sections. The accessory convention permits a connected torsor to become disconnected. The already independently verified result $a(A_7) = 60$ is taken as an input.

Question	Result and remaining limit
1. Upper bound	A direct twisting proof establishes the subgroup upper bound for every torsor, including disconnected base changes. The precise bibliographic convention is explained.
2. Full monodromy	$72 \leq \mu(A_7) \leq 120$. In fact $\mu(A_7) \in \{72, 84, 90, 96, 108, 120\}$. On each $(2, 4, 7)$ curve the minimum is either 90 or 180. The exact global value is not determined.
3. Other groups	$a(A_6) = 12$ is proved. Explicit general upper bounds are given for A_n and $\mathrm{PSL}_2(q)$; the induction formula does not by itself supply exact values for all parameters.
4. Towers	A fixed-point/Albanese extension of the correspondence argument is proved. A degree-two full- A_7 accessory already admits a smooth rational model with no fixed point. No unrestricted resolvent-degree lower bound follows.
5. Gonality	A stronger harmonic Hersch inequality removes the quartic-optimisation requirement. A complete sufficient certificate using an exact trial space is proved below. Its finite-element bounds have not been certified in this note.

Updated input. At the requested repository head, the gonality certificates cover all 70 A_7 -classes of the other ten rigid signatures and $(2, 4, 7)$ classes 12 and 14. Only classes 0 and 1 remain. Their certified inputs are

$$A = 540\pi, \quad a := 0.34089 \leq \lambda_1 \leq 0.36318, \quad \lambda' \geq b := 0.55998,$$

with E_1 one copy of $14_{(5,2)}$. The degree lattice in the earlier reference is the same result as the repository's Lemma 7.1. These facts replace the older reference's status statements. The reported 0.2% conformal ceiling is numerical; the underlying ceiling inequality is proved in the repository.

1 Upper bounds, linear models and disconnected torsors

Exact citation. Farb–Wolfson [2], footnote 4 immediately after equation (1.2) in Problem 1.4, states independence of the faithful linear model and cites Farb–Kisin–Wolfson [3], published Example 4.6 and Lemma 4.9. The accessible 2019 preprint has linear versality in Example 4.1.8(1) and a relative-dimension comparison in Lemma 4.1.12. Its cover category permits disconnected normal schemes and induced subgroup covers.

There is a qualification: that preprint's accessory classes are required to be closed under products; the class of single covers of degree at most n is not. Consequently, simply invoking its accessory-class lemma verbatim leaves a convention issue. For the curve upper bound used here, the following argument proves exactly what is needed without that invocation.

Proposition 1.1 (Universal subgroup upper bound). *Let $H \leq G$ and let C carry an H -linearized moving bundle L of degree m . For every field $F/\mathbb{C}$ and every G -torsor T/F , there is a finite field extension F'/F of degree at most $[G : H]m$ after which T descends to a field of transcendence degree at most one over $\mathbb{C}$. No connectedness hypothesis is imposed on $T_{F'}$.*

Proof. The finite etale scheme T/H has degree $[G : H]$. Choose a closed point with residue field F_0 ; then $[F_0 : F] \leq [G : H]$. It specifies an H -reduction P of T_{F_0} .

Twist C and L by P . The twist $C_P = P \times^H C$ is a geometrically connected smooth curve over F_0 . Genuine linearization descends L to an actual line bundle L_P of degree m , with a base-point-free space of sections of dimension $h^0(C, L) \geq 2$.

The field F_0 is infinite. A general F_0 -section of L_P has a reduced zero divisor of degree m , and can be chosen to avoid the finite locus corresponding to points of C with nontrivial stabilizer. Choose one closed point of this divisor and call its residue field F' . Then $[F' : F_0] \leq m$.

An F' -point of C_P is an H -equivariant map $P_{F'} \rightarrow C_{F'}$. Because its image lies in the free locus, $P_{F'}$ is the pullback of $C \rightarrow C/H$ at the corresponding F' -point of C/H . That point is defined over a subfield of F' of transcendence degree at most one. Induction from H to G gives the required descent of $T_{F'}$. A constant point gives a split torsor and transcendence degree zero. Finally, $[F' : F] \leq [G : H]m$. $\square$

For a connected G -extension L_0/K and an accessory F/K , put $L_0 \cap F = L_0^H$ after choosing an embedding. Then

$$\mathrm{Gal}(L_0 F/F) = H, \quad \mathrm{Spec}(L_0 \otimes_K F) \simeq \mathrm{Ind}_H^G \mathrm{Spec}(L_0 F).$$

Thus the base-changed G -torsor is generally disconnected, while its chosen component is a connected H -torsor. The induced curve $G \times^H C$ has a free generic G -action; this is stronger than faithfulness of a permutation action on components.

Combining Proposition 1.1 with the already verified correspondence lower bound gives, for the convention in Round 5,

$$a(G) = \min_{H \leq G} [G : H] \mu(H), \quad \mu(1) = 1.$$

In particular the Klein quartic for the index-15 subgroup $H = \mathrm{PSL}_2(7)$ supplies the degree-60 upper bound on every faithful linear model, and on every G -torsor. It does not supply a connected full- A_7 compression of degree 60.

2 Full-monodromy degree: $72 \leq \mu(A_7) \leq 120$

Write $N(C) = 2520/\mathrm{lcm}(e_i)$ for the linearized degree lattice. The exponent of A_7 is 420, so every genuinely linearized degree is divisible by 6. We use the verified input $\mu(A_7) \geq 60$ and minimum faithful genus 136.

Lemma 2.1 (Effective orbit obstruction). *If a finite subgroup H fixes a nonzero section of a linearized bundle of positive degree n on C , then n is a nonnegative integral combination of the sizes of H -orbits on C . In particular,*

$$n \geq \frac{|H|}{\max_{p \in C} |H \cap G_p|}.$$

Proof. The zero divisor of the fixed section is a nonzero effective H -invariant divisor. Its orbit decomposition proves both assertions. $\square$

Theorem 2.2. *There is no A_7 -linearized moving bundle of degree at most 71 on a faithful curve. Hence $\mu(A_7) \geq 72$.*

Proof. Suppose such a bundle exists. Normalize the image of its complete series, obtaining a faithful A_7 -curve Y with an A_7 -linearized bundle of degree $n \leq 71$. Faithfulness follows from simplicity: a nontrivial kernel would be all of A_7 , whose order would divide the degree of the map to the image, which is at most 71.

The earlier lower bound forces $n \geq 60$, and $6 \mid n$, so $n = 60$ or 66 . The genuine section representation has dimension at least 6, and Castelnuovo gives $g(Y) \leq \pi(n, 5)$. Exact Riemann–Hurwitz and lattice enumeration gives

n	$\pi(n, 5)$	possible signature	$g(Y)$	$N(Y)$
60	406	(2, 6, 7)	241	60
60	406	(3, 4, 7)	346	30
66	496	none	—	—

The short exact enumeration is reproduced in the appendix. Quotient genus at least one gives no additional cases.

For $n = 60$, $\pi(60, 8) = 220 < 241$, so the section dimension is at most 8. The only nontrivial irreducible A_7 -representation of dimension at most 8 is the standard six-dimensional representation. There are no invariant sections: their zero divisors would be effective A_7 -divisors of degree 60, whereas every orbit has size at least $2520/7 = 360$. Thus the section representation is exactly the standard 6.

Its restriction to a point-stabilizer A_6 contains a fixed vector. On a (2, 6, 7) curve, an inertia group meets this A_6 in order at most 3: an element of order 6 has cycle type (3)(2)(2), and a 7-cycle fixes no letter. Therefore every A_6 -orbit has size at least $360/3 = 120$. On a (3, 4, 7) curve the corresponding minimum is $360/4 = 90$. Both contradict the degree-60 fixed section by Lemma 2.1. $\square$

Theorem 2.3 (Explicit degree-120 construction). *The curve*

$$C_{120} = \left\{ [x_1 : \cdots : x_7] \in \mathbb{P}^6 : \sum_{j=1}^7 x_j^k = 0 \quad (k = 1, 2, 3, 4, 5) \right\}$$

is smooth and connected, has genus 481, and has a faithful A_7 -action. Its hyperplane bundle is genuinely A_7 -linearized, base-point-free, and has degree 120. Consequently $\mu(A_7) \leq 120$.

Proof. The seven distinct seventh roots of unity supply a point. At every point there are at least six distinct nonzero coordinate values. Indeed, if there were $k \leq 5$ such values $a_1, \dots, a_k$, with positive multiplicities m_j , the first k moment equations would give

$$\sum_j m_j a_j^r = 0 \quad (1 \leq r \leq k).$$

The coefficient matrix has determinant $(\prod_j a_j) \prod_{i < j} (a_j - a_i) \neq 0$, contradicting $m_j > 0$.

The Jacobian of the five defining equations consists, up to nonzero row factors, of the rows (1) , (x_j) , (x_j^2) , (x_j^3) and (x_j^4) . Vandermonde rank is five. Thus this is a smooth complete intersection. The Koszul resolution gives $H^0(C_{120}, \mathcal{O}) = \mathbb{C}$, hence connectedness; smoothness then gives irreducibility. Its degree is $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$. Adjunction gives $K_{C_{120}} = \mathcal{O}(8)$, hence $2g - 2 = 960$.

Coordinate permutations preserve the curve. A 3-cycle moves the seventh-root point projectively, since the unchanged nonzero coordinates force the projective scalar to be one. Simplicity makes the A_7 -action faithful. The permutation representation on $\mathbb{C}^7$ linearizes $\mathcal{O}(1)$; after the relation $\sum x_j = 0$, its sections include the standard 6. $\square$

Proposition 2.4 (A finite list of remaining values).

$$\mu(A_7) \in \{72, 84, 90, 96, 108, 120\}.$$

Proof. For a bundle attaining the global minimum, its normalized complete-series image has the same degree: otherwise its hyperplane bundle would realize a smaller value of $\mu(A_7)$. Therefore Castelnuovo and the lattice apply with $n = \mu(A_7)$ itself. They exclude 78 and 102: the appendix gives no compatible signature below $\pi(78, 5) = 703$ or $\pi(102, 5) = 1225$.

For $n = 114$, the only candidate is $(3, 4, 5, 7)$, with genus 1354. Since $\pi(114, 6) = 1221$, the sections must form the standard 6. In its seven-coordinate realization, $\prod_j x_j$ is a nonzero invariant section of degree $7n = 798$. Nonzero follows because an irreducible curve in the union of coordinate hyperplanes would lie in one, and transitivity would then force all coordinates to vanish. But the available orbit sizes are 840, 630, 504, 360 and 2520; no nonnegative combination equals 798. This excludes 114. Divisibility by 6 and Theorems 2.2–2.3 complete the list. $\square$

The genus-136 curves: two explicit degree-90 tests

Fix any $(2, 4, 7)$ curve. Let D_e be the reduced divisor above the branch point of order e , and let F denote a full quotient fibre. Then

$$\deg(D_2, D_4, D_7) = (1260, 630, 360), \quad 2D_2 \sim 4D_4 \sim 7D_7 \sim F.$$

All these bundles have canonical A_7 -linearizations. Put

$$B = 2D_7 - D_4, \quad \deg B = 90; \quad T = D_2 - 2D_4, \quad 2T = 0; \quad 2B \sim 2D_4 - 3D_7, \quad \deg(2B) = 180.$$

Riemann–Roch gives $h^0(2B) \geq 180 - 136 + 1 = 45$. Its fixed divisor is A_7 -invariant and has degree at most 180, hence is zero since the minimum orbit size is 360. Therefore $2B$ is already a linearized moving bundle.

Every linearized class has an invariant rational divisor. Since A_7 has no characters and the quotient is $\mathbb{P}^1$, the equivariant Picard group has presentation

$$\mathrm{Pic}_{\mathrm{lin}}^{A_7}(C) = \langle D_2, D_4, D_7 \mid 2D_2 = 4D_4 = 7D_7 \rangle \simeq \mathbb{Z} \oplus \mathbb{Z}/2.$$

Thus B and $B + T$ are exactly the two degree-90 classes; also $K_C = 3(B + T)$. Any nonzero space of sections of a degree-90 class has dimension at least 6, since an invariant section is impossible. Consequently, on this fixed curve,

$$\mu_C(A_7) = \begin{cases} 90, & h^0(B) > 0 \text{ or } h^0(B + T) > 0, \\ 180, & h^0(B) = h^0(B + T) = 0. \end{cases}$$

Computing these two spaces of sections is a concrete unresolved test. Neither the degree lattice nor Riemann–Roch at degree 90 decides it.

3 What induction computes beyond A_7

The formula is an exact reduction to linearized curve geometry, rather than a closed formula for the two infinite families. Equivalently,

$$a(G) = \frac{|G|}{\max_{H \leq G} (|H|/\mu(H))}.$$

The optimum balances subgroup size against moving degree; genus alone does not decide it. The representation, lattice and effective-orbit tests must still be supplied for each relevant subgroup.

Proposition 3.1.

$$a(A_5) = 2, \quad a(A_6) = 12, \quad a(\mathrm{PSL}_2(7)) = 4.$$

Proof. The icosahedral action on $\mathbb{P}^1$ has the genuinely linearized tangent bundle $\mathcal{O}(2)$. Degree one would require a faithful projective action lifted to an honest two-dimensional representation of A_5 , which does not exist. Thus $\mu(A_5) = 2$; all proper subgroup indices are at least 5.

For A_6 , the minimum nontrivial representation dimension is 5 and the exponent is 60. The lattice forces $6 \mid n$. If a moving degree were at most 11, its faithful normalized image would have $n \leq 6$ and genus at most $\pi(6, 4) = 2$. There is no faithful genus-zero or genus-one A_6 action; Hurwitz also excludes genus two. Hence $\mu(A_6) \geq 12$.

The proper indices below 12 that divide 360 are 2, 3, 4, 5, 6, 8, 9, 10. Indices below 6 are impossible by simplicity and the order bound for a faithful permutation representation. An index-6 action embeds A_6 as the index-two alternating subgroup of S_6 , so its stabilizer is A_5 . Subgroups of orders 45 or 40 have a normal Sylow 5-subgroup, hence would lie in its normalizer, of order 10 in A_6 ; this excludes indices 8 and 9. A subgroup of order 36 cannot act faithfully on $\mathbb{P}^1$: the finite Möbius groups of that order are cyclic or dihedral and would require an element of order at least 18, absent from A_6 . Its moving degree is therefore at least 2. The induction minimum is consequently 12, attained by $[A_6 : A_5]\mu(A_5) = 6 \cdot 2$.

For $\mathrm{PSL}_2(7)$, a degree at most 3 would give a faithful curve of genus at most $\pi(3, 2) = 1$, impossible. The Klein quartic gives $\mu = 4$, and the minimum proper subgroup index is 7. $\square$

The power-sum construction in Theorem 2.3, with n coordinates and equations of degrees $1, \dots, n-2$, gives for $n \geq 5$

$$\mu(A_n) \leq (n-2)!, \quad a(A_n) \leq (n-2)!.$$

Induction from A_7 also gives, for $n \geq 7$,

$$a(A_n) \leq \min \left\{ (n-2)!, \frac{n!}{84} \right\}.$$

For a cyclic group and an odd dihedral group D_{2m} , $\mu = 1$: their faithful actions on $\mathbb{P}^1$ lift to projectively faithful honest two-dimensional representations. In particular, the torus normalizers in $G = \mathrm{PSL}_2(q)$ give

$$a(G) \leq \begin{cases} q(q-1)/2, & q \text{ even}, \\ q(q-1)/2, & q \equiv 1 \pmod{4}, \\ q(q+1)/2, & q \equiv 3 \pmod{4}. \end{cases}$$

For odd q , select the normalizer whose torus order $(q \pm 1)/2$ is odd; for even q , use the torus of order $q+1$. These are upper bounds, often loose in small cases. They do not establish an exact formula for general q or a theorem that the optimum always has small genus.

4 A tower analogue and a fixed-point failure

Proposition 4.1 (Correspondences over more general bases). *Let B be a smooth connected projective faithful G -variety with a G -fixed point. Suppose every morphism $B \rightarrow \text{Pic}^e(C)$ is constant for every faithful G -curve C and every integer e . If an integral equivariant degree- d cover $W \dashrightarrow B$ dominates such a curve C , then $d \geq \mu(G)$. In particular the hypothesis holds when $\text{Alb}(B) = 0$, including rationally connected B .*

Proof. The joint-image divisor $Z \subset B \times C$ is integral, invariant and Cartier, and has degree $e \mid d$ over B . Its fibre classes define a morphism $B \rightarrow \text{Pic}^e(C)$, hence have one constant underlying class. Restrict $\mathcal{O}(Z)$ to the fixed-point fibre to obtain a genuine G -linearization of that class. Since Z dominates C , its general fibre divisors cannot all be the same divisor. The class therefore has at least two sections. Removing the invariant fixed part gives a moving bundle of degree at most $e \leq d$. $\square$

This needs constancy of the Picard family, not the stronger equality $\text{Pic}(B \times C) = \text{Pic}(C)$. If the class family is nonconstant, it supplies an Albanese direction mapping to the Jacobian of C . If it is constant, the fixed point is what produces genuine linearization. These identify two separate ways the mechanism can fail on an intermediate model.

Proposition 4.2 (Failure already after a quadratic accessory). *There is a degree-two accessory of a generic A_7 -torsor that retains full monodromy but has a smooth rational equivariant model with no A_7 -fixed point.*

Proof. Let V be the standard 6 and q its nondegenerate invariant quadratic form. The integral cover

$$W = \{(v, t) \in V \oplus \mathbb{C} : t^2 = q(v)\} \longrightarrow V$$

has degree two, with A_7 acting trivially on t . Its only fixed point is its singular vertex. Blow up that vertex. The resulting smooth variety is the total space of $\mathcal{O}_Q(-1)$ over the smooth five-dimensional quadric

$$Q = \{[v : t] \in \mathbb{P}(V \oplus \mathbb{C}) : t^2 = q(v)\}.$$

The only invariant line in $V \oplus \mathbb{C}$ is the t -axis, which does not lie on Q . Thus Q and the blow-up have no fixed point. The quadric and its line-bundle total space are rational. One can also compactify to the smooth rational projective bundle $\mathbb{P}(\mathcal{O}_Q \oplus \mathcal{O}_Q(-1))$, still with no fixed point.

The extension of the invariant field has degree two. Its intersection with $\mathbb{C}(V)$ is the original invariant field, because a degree-two subextension of an A_7 -Galois extension would give a quotient of order two. Therefore the base-changed torsor remains connected with group A_7 . $\square$

For any accessory tower with step degrees $d_1, \dots, d_r$ that eventually gives a curve compression, the composite is one accessory. Hence

$$\prod_i d_i \geq 60; \quad \prod_i d_i \geq 72 \quad \text{if connected full } A_7 \text{ monodromy is retained.}$$

These product bounds impose no bound on the number of steps and do not control the essential dimensions of the individual covers. Proposition 4.1 gives a useful conditional obstruction on intermediate bases, but Proposition 4.2 prevents assuming that its hypotheses persist. The unrestricted question $\text{RD}(A_7) > 1$ remains unresolved here.

5 Gonality: a stronger harmonic Hersch inequality

Use the positive Laplacian. For real vector fields u, v, w on the surface put

$$\Theta(u, v, w) = \frac{1}{2} \sum_{ijk} \epsilon_{ijk} \int_C u_i dv_j \wedge dw_k.$$

Stokes makes Θ totally symmetric, and $T(u) = \Theta(u, u, u)$. If $x : C \rightarrow S^2$ is holomorphic of degree m , then

$$E(x) := \int_C |\nabla x|^2 = 8\pi m, \quad \Delta x = |\nabla x|^2 x.$$

After a Mobius transformation it is balanced: $\int_C x = 0$. Write $x = y + z$ with $y \in E_1 \otimes \mathbb{R}^3$, and set

$$s = \|z\|_{L^2}^2, \quad t = \|y\|_{L^2}^2 = A - s, \quad \varepsilon = 8\pi m - \lambda_1 A, \quad E(z) = \varepsilon + \lambda_1 s.$$

The certified representation condition gives $T(y) = 0$ and makes $\beta(\wedge^2 E_1)$ orthogonal to $1 \oplus E_1$.

Keep the repository's definitions

$$Q^*(\omega) = \langle \beta(\omega), (\Delta - \lambda_1)^{-1} \beta(\omega) \rangle, \quad \kappa^2 = \sup_{\sum_a |w_a|^2 = 1} \sum_{\text{cyc}} Q^*(w_a \wedge w_b),$$

and $\Lambda = \sup_p \lambda_{\max} \sum_i \nabla \varphi_i(p) \otimes \nabla \varphi_i(p)$ for an L^2 -orthonormal eigenbasis. Put $\Gamma = \sqrt{A\Lambda}$.

Theorem 5.1 (Harmonic Hersch). *Every such holomorphic map satisfies*

$$\boxed{\lambda_1(A - s) \leq 4\kappa\sqrt{\varepsilon}(A - s) + \sqrt{\Lambda(A - s)s(\varepsilon + \lambda_1 s)}} \quad (\text{HH})$$

Proof. Conformality and the harmonic-map equation give the exact identity

$$\Theta(x, x, y) = \frac{1}{2} \int_C |\nabla x|^2 x \cdot y = \frac{1}{2} \langle \nabla x, \nabla y \rangle = \frac{\lambda_1 t}{2}.$$

On the other hand, total symmetry and $T(y) = 0$ give

$$\Theta(x, x, y) = 2\Theta(z, y, y) + \Theta(z, y, z).$$

The first term obeys $|\Theta(z, y, y)| \leq \kappa\sqrt{\varepsilon}t$ by Cauchy–Schwarz in the positive form $E(z) - \lambda_1 \|z\|^2$. For the second, the cross-product operator estimate used in the repository gives

$$|2\Theta(z, y, z)| \leq \sqrt{\Lambda t} \sqrt{sE(z)}.$$

Multiply the identity by two and apply these bounds. □

The additional ingredient is harmonicity of the *actual map* x . It replaces a cubic remainder estimate by an exact energy pairing. Thus (HH) is stronger than the original inequality for the holomorphic maps relevant to gonality.

A wide sufficient box, with no quartic optimisation

Let $B_* := \|\beta^*(\Delta - \lambda_1)^{-1} \beta\|_{\wedge^2 E_1 \rightarrow \wedge^2 E_1}$. For three vectors with total squared norm one,

$$\sum_{a < b} \|w_a \wedge w_b\|^2 \leq \sum_{a < b} |w_a|^2 |w_b|^2 \leq \frac{1}{3}.$$

Therefore $\kappa^2 \leq B_*/3$ rigorously; no global maximum on S^{41} is required.

Corollary 5.2. *For a class 0 or 1 curve with the accepted eigenvalue certificates, the two inequalities*

$$\boxed{B_* \leq 3 \cdot 10^{-4}, \quad \Gamma \leq 2}$$

would certify $\text{gon}(C) \geq 25$.

Proof. It suffices to exclude $m \leq 24$. Put $\tau = 16/45 = 8\pi \cdot 24/A$. Li-Yau already excludes the case $\lambda_1 > \tau$. For $\lambda = \lambda_1 \in [a, \tau]$, the spectral gap gives

$$\frac{s}{A} \leq u := \frac{\tau - \lambda}{b - \lambda} < 1, \quad \frac{E(z)}{A} \leq bu.$$

Dividing (HH) by $A - s$ yields the necessary inequality

$$\lambda \leq 4\sqrt{B_*/3}\sqrt{A(\tau - \lambda)} + \Gamma\sqrt{b} \frac{\tau - \lambda}{\sqrt{(b - \lambda)(b - \tau)}}. \quad (5.1)$$

Both terms on the right decrease as λ increases. At $\lambda = a$, the stated box bounds the right side by $0.30324 < a = 0.34089$. The contradiction is uniform; no sampling in λ is used. The rational square-root checks are in the appendix. $\square$

At the repository's *numerical* $B_* \simeq 1.2769 \cdot 10^{-4}$ and $\Gamma \simeq 1.70$, the common inflation allowed for $\sqrt{B_*/3}$ and Γ is about 1.56. This is a planning comparison, not an enclosure of the true constants. In particular the optimisation output 0.6143 is unnecessary for this route and is not being treated as a certified global maximum.

A sufficient certificate on an exact trial space

The most practical route is to certify a trial-space version of (HH). It avoids pointwise bounds for the *true* eigenfunctions. Choose an exact G -stable finite-element space E_h isomorphic to one copy of $14_{(5,2)}$, contained in the mean-zero functions. The cubic identities remain valid for continuous piecewise smooth fields: cellwise Stokes has cancelling edge traces. The true L^2 and Dirichlet forms are invariant; Schur's lemma gives a scalar λ_h such that

$$\langle \nabla u, \nabla v \rangle = \lambda_h \langle u, v \rangle \quad (u, v \in E_h).$$

Define its residual bound ρ by

$$|\langle \nabla u, \nabla v \rangle - \lambda_h \langle u, v \rangle| \leq \rho \|u\|_{L^2} \|\nabla v\|_{L^2} \quad (u \in E_h, v \perp 1).$$

For an exact L^2 -orthonormal basis ψ_i of E_h , define β_h , Λ_h and $\Gamma_h = \sqrt{A\Lambda_h}$ by the same formulas as above. Let B_h bound the operator

$$\beta_h^*(\Delta - r)^{-1}\beta_h, \quad r = 0.351.$$

Here the inverse is taken on the four real isotypes in

$$\wedge^2 14_{(5,2)} = (10 \oplus \overline{10}) \oplus 15 \oplus 21 \oplus 35.$$

They all have spectrum at least b . This decomposition has no 1 or 14, so the brackets are automatically orthogonal to E_h and to the true first eigenspace.

Theorem 5.3 (A complete trial-space target). *The following four certified bounds, separately for classes 0 and 1, suffice for $\text{gon}(C) \geq 25$:*

$$\boxed{\lambda_h \leq 0.351, \quad \rho \leq 0.005, \quad B_h \leq 2 \cdot 10^{-4}, \quad \Gamma_h \leq 1.9.} \quad (5.2)$$

Only the accepted spectral bounds $\lambda_1 \geq a$ and $\lambda' \geq b$ are needed in addition.

Proof. Let P and P_h be the L^2 projections onto E_1 and E_h . The spectral expansion of the residual of a unit $u \in E_h$ has squared energy-dual norm

$$\sum_{\mu > 0} \frac{(\mu - \lambda_h)^2}{\mu} |u_\mu|^2 \leq \rho^2.$$

For $\mu \geq b$ the coefficient is at least $(b - r)^2/b$. Equal ranks then give

$$\|P - P_h\| \leq \delta := \frac{\rho\sqrt{b}}{b - r}, \quad E(z) \geq \nu \|z\|^2 \quad (z \perp 1 \oplus E_h), \quad \nu := b - (b - a)\delta^2.$$

The last inequality follows by splitting z into its true first-eigenspace component, of squared norm at most $\delta^2 \|z\|^2$, and its complement. The targets imply $\nu > 0.55990 > r$.

Now project a balanced degree- $m \leq 24$ map onto $E_h \otimes \mathbb{R}^3$, writing $x = y + z$, $t = \|y\|^2$, $s = \|z\|^2$, $E_z = E(z)$ and $c = \langle \nabla y, \nabla z \rangle$. Then

$$t + s = A, \quad |c| \leq \rho\sqrt{tE_z}, \quad E(x) = \lambda_h t + 2c + E_z \leq A\tau.$$

Since $\lambda_h \geq a$ and $s \leq E_z/\nu$, putting $R_z = \sqrt{E_z/A}$ gives

$$(1 - a/\nu)R_z^2 - 2\rho R_z \leq d := \tau - a, \quad R_z \leq R := \frac{\rho + \sqrt{\rho^2 + (1 - a/\nu)d}}{1 - a/\nu}.$$

The targets imply $R < 0.20685$.

Harmonicity now gives the exact trial-space identity

$$\lambda_h t + c = 4\Theta(z, y, y) + 2\Theta(z, y, z),$$

because $T(y) = 0$ still follows from the representation. The form $E(z) - \lambda_h s$ is positive on $z \perp 1 \oplus E_h$. Since $\lambda_h \leq r$ and the brackets lie in the other isotypes, its dual bracket bound is at most B_h . Therefore, with $k = \sqrt{B_h/3}$,

$$\lambda_h t + c \leq 4k\sqrt{E_z - \lambda_h s}t + \sqrt{\Lambda_h t s E_z}.$$

Moreover $E_z - \lambda_h s = E(x) - \lambda_h A - 2c \leq A(d + 2\rho R)$. Combining all the upper and lower bounds gives the necessary inequality

$$a(1 - R^2/\nu) - \rho R \leq 4k\sqrt{A}\sqrt{d + 2\rho R} + \Gamma_h R^2/\sqrt{\nu}. \quad (5.3)$$

Under (5.2), its left side exceeds 0.3138 and its right side is less than 0.2827, as checked by rational bounds in the appendix. This is impossible. $\square$

How to produce the four verified bounds

Exact symmetry first. Start with an S_5 -invariant dyadic finite-element trial function on C/S_5 . Its translate span is a quotient of $\text{Ind}_{S_5}^{A_7} 1 = 1 + 6 + 14_{(5,2)}$. Apply the exact central character projector for 14 and certify nonzero norm. Its span is then one exact copy of 14, independent of floating eigensolver errors. Normalize using a verified enclosure of the true mass Gram matrix.

Invariant forms make λ_h and the residual Gram scalar on this copy. Higher-order continuous elements on this small quotient can reduce the residual cost; the coarse P1 residual is not presumed to meet (5.2).

Residual by equilibrated flux. For a unit $u \in E_h$, a flux p with $-\operatorname{div} p = \lambda_h u$ gives

$$R_u(v) = \int_C (\nabla u - p) \cdot \nabla v, \quad \|R_u\|_{H_E^{-1}} \leq \|\nabla u - p\|_{L^2}.$$

Use an $H(\operatorname{div})$ flux with matching normal traces on glued edges. If coefficient approximation leaves a divergence defect f , include its energy-dual error, bounded by $\|f\|_{L^2}/\sqrt{a}$. Interval quadrature, flux mismatch and assembly errors all enter the final ρ ; an unverified floating residual does not suffice.

Four scalar resolvent bounds. The bracket form is invariant and the real wedge-square decomposition is multiplicity-free, so B_h is the maximum of four scalars. For a unit wedge vector in one isotype, put $f = \beta_h(\omega)$ and

$$q_r(u, v) = \langle \nabla u, \nabla v \rangle - r \langle u, v \rangle.$$

For any approximate solution u_h in that isotype, with residual $\mathcal{R}(v) = \langle f, v \rangle - q_r(u_h, v)$, the exact identity is

$$\langle f, (\Delta - r)^{-1} f \rangle = 2 \langle f, u_h \rangle - q_r(u_h, u_h) + \|\mathcal{R}\|_{q_r^{-1}}^2.$$

If $|\mathcal{R}(v)| \leq \eta \|\nabla v\|$, coercivity $q_r(v, v) \geq (1 - r/b)E(v)$ gives the rigorous upper bound

$$\boxed{2 \langle f, u_h \rangle - q_r(u_h, u_h) + \frac{\eta^2}{1 - r/b}}. \quad (5.4)$$

This is a complementary-energy upper bound, not the Rayleigh–Ritz lower bound. Equilibrated fluxes certify η ; uncertainty in f must be added to it and to the pairing enclosure. On P1 elements the load density $\beta_h(\omega) dA$ is an exact coordinate Jacobian, which simplifies the load integration. A single verified unit vector per isotype suffices once exact equivariance and normalization have been established.

Gradient supremum on whole cells. In reference coordinates, let $M^{-1} = A_R/w_R$ be the inverse metric and $S_{ab} = \sum_i \partial_a \psi_i \partial_b \psi_i$. If $M^{-1} = LL^T$ is a lower Cholesky factorization, the physical frame matrix is

$$L^T S L, \quad \text{not } L S L^T.$$

The latter is used in `topo_hersch.py` at the requested head. Its contraction should be replaced by

```
Sm = np.einsum("eai,teab,ejb->teij", Lc, S, Lc)
```

The exact check is $\operatorname{tr}(L^T S L) = \sum_{ab} (M^{-1})_{ab} S_{ab}$. In the recovered coarse $n = 4$, class-0 calculation, the correction changed the sampled Γ_h from 1.70065 to 1.67134 and removed the trace mismatch. These are numerical mesh values, not certified suprema.

Even for P1 functions, only S is cellwise constant; the hyperbolic metric varies within the cell. Enclose M^{-1} over every whole reference cell and bound the largest eigenvalue of $L^T S L$, or use

$$\lambda_{\max} = \frac{\operatorname{tr}(M^{-1} S) + \sqrt{\operatorname{tr}(M^{-1} S)^2 - 4 \det(M^{-1}) \det S}}{2}.$$

A barycentre sample cannot certify Γ_h . No C1 enclosure of the true eigenfunctions is needed for Theorem 5.3.

What is still missing. The four numerical inequalities (5.2) have not been established with outward-rounded flux and metric bounds in this note. The theorem proves their sufficiency and supplies an implementable upper-bound formula. Accordingly, neither the remaining two gonality cases nor the global assertion $\gamma(A_7) \geq 25$ is being declared certified here.

A Short exact checks

For the signature enumeration, $\pi(n, r) = (r-1)\binom{k}{2} + kt$ where $n-1 = (r-1)k + t$. The following uses only exact rational arithmetic. The branch count limit 6 and quotient genus limit 1 suffice for the degrees listed: seven branch points have genus at least 1891, and quotient genus at least two has genus at least 2521, both beyond the largest Castelnuovo bound 1540.

```
from fractions import Fraction as F
from itertools import combinations_with_replacement as cr
from math import lcm, comb
def pi(n,r):
    k,t = divmod(n-1,r-1)
    return (r-1)*comb(k,2)+k*t
for n in (60,66,78,102,114):
    out=[]
    for h in (0,1):
        for count in range(1,7):
            for es in cr(range(2,8),count):
                g=1+1260*(2*h-2+sum(1-F(1,e) for e in es))
                N=2520//lcm(*es)
                if g.denominator==1 and 136<=g<=pi(n,5) and n%N==0:
                    out.append((es,h,int(g),N))
    print(n,pi(n,5),out)
# 60 406: (2,6,7), g=241,N=60; (3,4,7),g=346,N=30
# 66 496: none; 78 703: none; 102 1225: none
# 114 1540: (3,4,5,7),g=1354,N=6
```

Here are reproducible rational upper bounds for the two spectral contradictions. All decimals in the assertions are exact rationals; $\pi < 3.141593$ is the only transcendental upper bound used.

```
from fractions import Fraction as F
a,b,r,rho = map(F,('0.34089','0.55998','0.351','0.005'))
tau=F(16,45); Aup=540*F('3.141593'); d=tau-a
# Corollary 5.2: each displayed square bounds its radical.
assert 16*F('0.01')**2*Aup*d < F('0.19952')**2
assert 4*b*d*d/((b-a)*(b-tau)) < F('0.10372')**2
assert F('0.19952')+F('0.10372') < a
# Trial-space theorem: a rational upper root Rup.
delta2=rho*rho*b/(b-r)**2
nu=b-(b-a)*delta2
R=F('0.20685'); c=1-a/nu
assert nu>F('0.55990') and c*R*R-2*rho*R>d
lhs=a*(1-R*R/nu)-rho*R
assert lhs>F('0.3138')
assert 16*(F('0.0002')/3)*Aup*(d+2*rho*R)<F('0.17403')**2
assert F('1.9')**2*R**4/nu<F('0.10865')**2
assert F('0.17403')+F('0.10865')<F('0.2827')<F('0.3138')
```

References

- [1] A₇ project repository, [hilbert13/ladder](#), pinned head 1ef3f6f. Inputs: QUESTIONS_FOR_GPT.md, Round 5; ACCESSORY_60.md; VERIFICATION.md; FRAMEWORK_TOPOLOGICAL_HERSCH.md; numerical scripts and outputs cited in the text. These are upstream inputs, not a claim that every large certificate was rerun here.
- [2] B. Farb and J. Wolfson, *Essential dimension relative to branched covers of degree at most n*, [arXiv:2510.22786](#); [author PDF](#), especially Problem 1.4 and footnote 4.
- [3] B. Farb, M. Kisin and J. Wolfson, with an appendix by N. Harman, *Modular functions and resolvent problems*, Math. Ann. **386** (2023), 113–150, [DOI:10.1007/s00208-022-02395-8](#). Accessible [2019 preprint](#), Section 4.1. Numbering differs from the published citation in Farb–Wolfson’s footnote.